

Ling FENG

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EDUCATION

Shanghai Ocean University (Double First-Class University)

Shanghai, China

BEng Robotics

09.2021-06.2025

- Major: Robotics Engineering
- Average Points: First Class
- Related Courses: Computer Control Technology (97), Circuit Principles (93), Robot Structure Design Course Design (91), Complex Function and Integral Transformation (96), Signal Analysis and Processing (92), Course Design of Machine Vision and Sensor Technology (Excellent Grade).

University of Manchester

Manchester, UK

Msc Robotics

09.2025-06.2026(Expected)

- Expected Class: Distinction
- Related Courses:Autonomous Mobile Robots, Robotic Systems Design Project, Reinforcement Learning, Cognitive Robotics and Computer Vision, Robotic Systems, Robotic Manipulators

SKILLS

- **Software:** Python, C, ROS2, MATLAB, OpenCV, LabVIEW, Linux
- **Hardware:** Altium Designer, PCB Design, Raspberry Pi, Multisim, Keil
- **Modeling & Control:** SolidWorks, CAD, Control System, KukasimPro

WORK EXPERIENCE

Shanghai Quicktron Intelligent Technology Co., Ltd.

Position: Software Test Engineer

06.2025-09.2025

- Utilized tools such as QS and VOS to perform AGV module interaction testing, functional verification, and system deployment monitoring. Accurately identified product defects and conducted requirement analysis.
- Developed Python automation scripts, combined with Postman for API testing and SQL/Navicat for data validation, significantly improving test coverage and execution efficiency.
- Monitored logs and located issues via FinalShell, managed module firmware burning and upgrades. Collaborated with teams to drive continuous product quality optimization.

Suzhou Huibo Robot Co., Ltd.

Position: Robotics Engineer

06.2024-09.2024

- Programmed and debugged robots using teach pendants to achieve mobile robot positioning and navigation, as well as robotic arm pick-and-place operations.
- Conducted parameter configuration and programming for 2D smart cameras to enable automated image recognition.

PROJECTS

An Optimal-Path-Planning Method for Unmanned Surface Vehicles Based on a Novel Group Intelligence Algorithm (The Second Author of a Tier-1 SCI Journal)

2024

Position: Researcher

- Evaluated and developed path-planning algorithms for USVs, integrating a novel collective intelligence approach to address dynamic obstacles and water currents, ensuring efficient and safe navigation.
- Leveraged MATLAB and artificial bee colony algorithms for simulation and optimization, resulting in an 8% increase in path length, a 50% reduction in optimization time, and near 100% dynamic obstacle avoidance rate.

Research on Machine Vision-Based Water Surface Search and Rescue Object Detection Technology

Position: Author

12.2024-05.2025

- Implemented the YOLOv5 deep learning model for water surface target detection, achieving a 90% mAP in complex environments and ensuring outstanding recognition accuracy and real-time performance.
- Employed binocular camera calibration and stereo matching algorithms to generate high-precision depth maps, establishing a reliable 3D data foundation for distance measurement.
- Integrated detection outputs from YOLOv5 with depth map data, enabling accurate calculation of spatial coordinates and actual distances.

Innovation and Entrepreneurship Training Program for College Students-- Artificial Intelligence Fish Recognition

Position: Project Leader

09.2023-06.2024

- Utilized LabelImg for data annotation and YOLO for object detection, training the model for 30 epochs to achieve 93.33% validation accuracy with rapid convergence.
- Developed and trained AI models using Python and a large dataset of fish images, enabling robust learning and recognition capabilities across diverse fish species and environmental conditions. Created a GUI interface in PyCharm for visualizing fish image recognition results, featuring functionalities for selecting images, displaying predicted and actual categories, and exiting, to validate the system's performance.

51 Microcontroller Based Digital Clock with Calibration

Position: Independent Developer

03.2023

- Developed the core timekeeping logic and display driver in C language using the Keil IDE, implementing a precise timer interrupt service routine for the 51 Microcontroller.
- Programmed real-time calibration features with C, utilizing tactile button interrupts to modify time variables (hours, minutes, seconds), ensuring responsive and accurate clock adjustment.

Robotic Systems Design Project-Autonomous Object Retrieval Robot

Position: Project Leader

10.2025-04.2026(expected)

- Deployed and optimized YOLOv8 in ROS2 for real-time object detection and color classification using RGBD camera streams.
- Bridged perception and action by publishing object coordinates via ROS2, enabling closed-loop vision-guided grasping with the robotic arm.
- Utilized LiDAR and IMU data within the ROS2 navigation stack, developing and testing SLAM and path planning algorithms in Gazebo to enable reliable autonomous movement to target objects..

EXTRACURRICULAR ACTIVITIES

Cyber Security Summer Camp

Position: Excellent Camper

07.2023

- Participated in the Cybersecurity Summer Camp at Guangzhou University, focusing on advancing the practical and innovative capabilities in the field of cyberspace security. Engaged with top-tier academic and industry experts to deepen understanding of cutting-edge defensive techniques and fostered a proactive security mindset.
- Immersed in the advanced curriculum of cybersecurity, broadened technical horizons, and mastered knowledge of professional courses. The intensive 10-day program facilitated the acquisition of in-depth skills in network attack and defense, enhancing the ability to address real-world security challenges.

HONORS

- The Second Prize in the National English Competition for College Students (National Level) *04.2024*
- The First prize of People's Scholarship (University Level) *09.2022*
- The Third Prize of People's Scholarship (Twice) (University Level) *03.2023/09.2023*
- The School Piano Competition Excellence Award (University Level) *12.2023*
- The Outstanding Student Title(University Level) *09.2022*
- The Fifth Place in the Table Tennis Competition of the Campus Cup (University Level) *12.2022*
- The Social Work Activist Designation (University Level) *09.2023*

ADDITIONAL INFORMATION

- **Certificate:**Level 3 AI Trainer Certification (Python Focus), IELTS 6.5 (7 for speaking) , Piano Level-10
- **Hobbies:** Piano, Swimming, Table tennis, Travel, Photography.